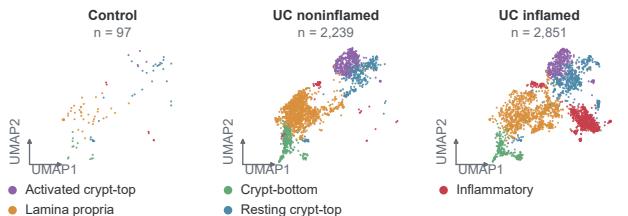
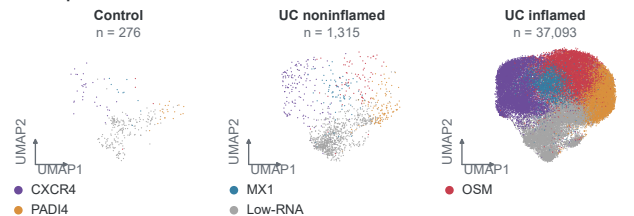


a Human UC fibroblast and neutrophil state landscape

Fibroblasts



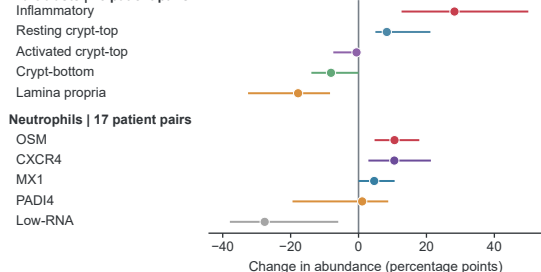
Neutrophils



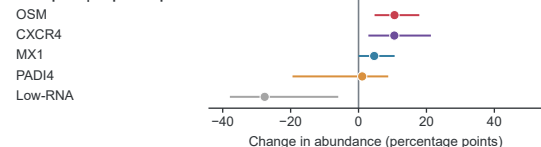
b Paired cell-state remodeling

Inflamed minus noninflamed; median and 95% CI

Fibroblasts | 13 patient pairs

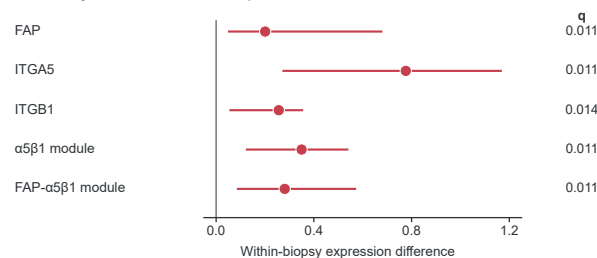


Neutrophils | 17 patient pairs



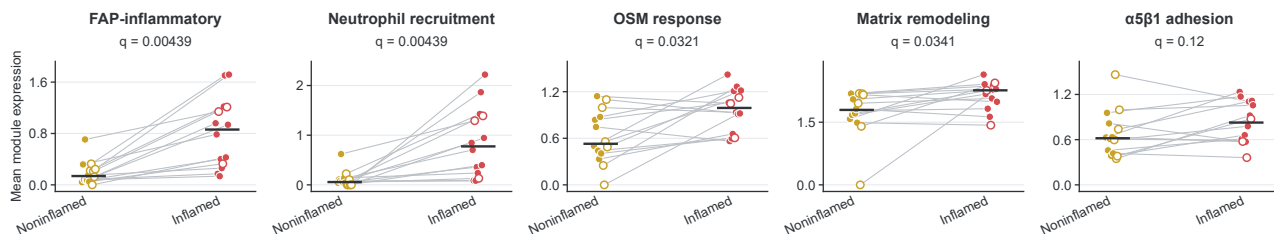
c Inflammatory-fibroblast activation

Inflammatory vs other fibroblasts; 10 biopsies



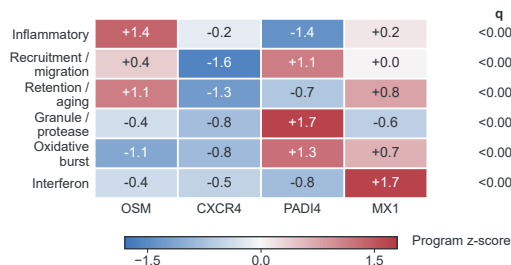
d Paired fibroblast program changes

14 patient pairs; lines connect patients; black bars, medians; open points, <10 recovered cells



e Neutrophil state-associated RNA programs

11 matched patients; ≥20 cells/state; naming genes excluded



f Predicted reciprocal signals

Fibroblast → neutrophil

HGF → CD44

CXCL1 → CXCR2

CSF3 → CSF3R

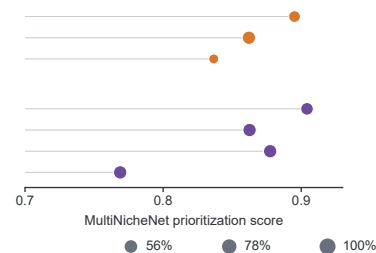
Neutrophil → fibroblast

OSM → OSMR

IL1B → IL1R1

NAMPT → ITGA5

NAMPT → ITGB1



Equal patient weights; Friedman q (BH, 9 programs).

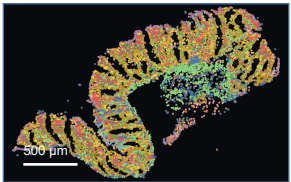
Paired coexpression: Extended Data Fig. 1c.

a CODEX cell types and neighborhoods

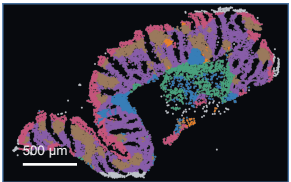
24 independent patients; matched fields and coordinates

Cell types

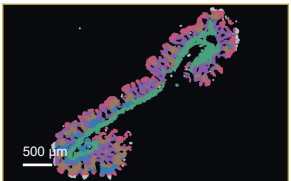
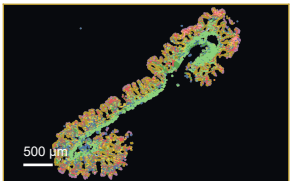
Control | P06



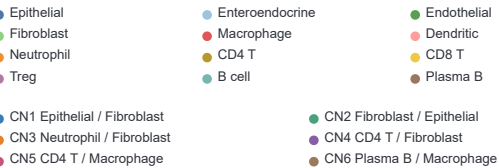
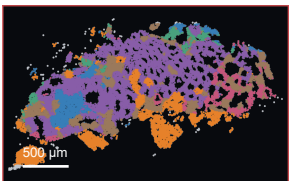
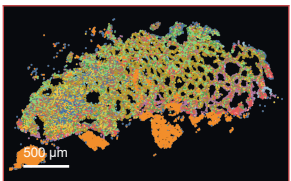
Cellular neighborhoods (50 μm)



UC noninflamed | P17

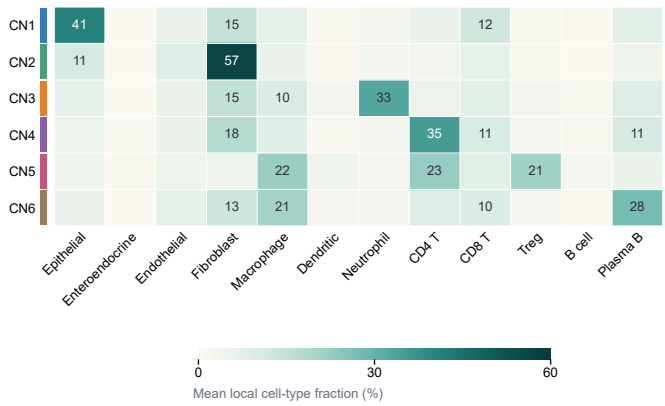


UC inflamed | P12



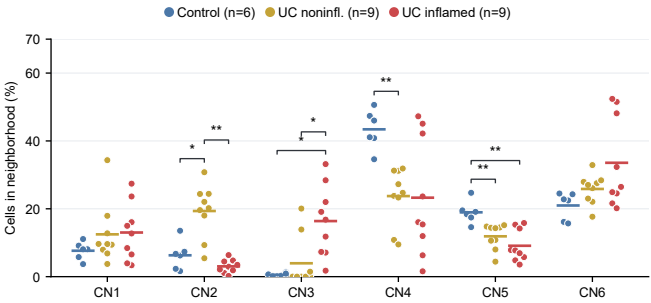
b Neighborhood composition

Patient-averaged local cell-type composition



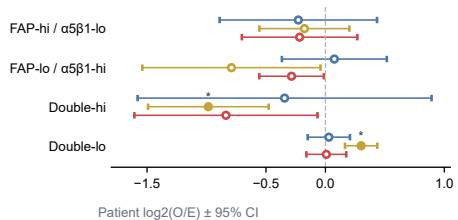
c Tissue neighborhood occupancy

Patient values and group means



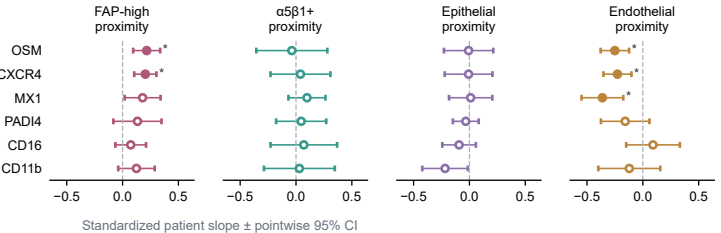
d Availability-adjusted proximity

50-μm proximity relative to a stratified spatial null



e Cell proximity and neutrophil markers

Adjusted associations in 8 patients with inflamed UC



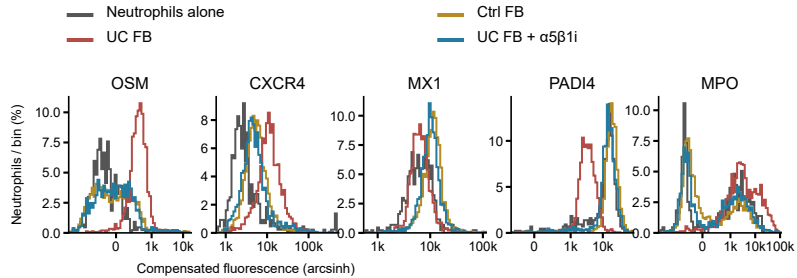
a Culture and blockade design

Peripheral-blood neutrophils

Alone
Untreated / NAMPTi / α5β1i
Ctrl FB
Untreated
UC FB
Untreated / NAMPTi / α5β1i / dual

Independent preculture reference; RNA profiling

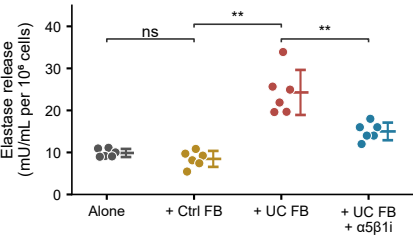
b Neutrophil protein distributions



c Fibroblast exposure and blockade effects on neutrophil proteins

	Fibroblast exposure			Neutrophils alone		Blockade in UC FB coculture				Δ protein z score
	Ctrl FB vs alone	UC FB vs alone	UC FB vs Ctrl FB	NAMPTi vs untreated	α5β1i vs untreated	NAMPTi vs untreated	α5β1i vs untreated	Dual vs untreated	Dual vs NAMPTi	
OSM	+0.07	+1.87**	+1.80**	+0.03	-0.19	-0.01	-1.96**	-1.85**	-1.84*	
CXCR4	+0.83**	+2.86**	+2.03**	-0.04	+0.65**	-1.09*	-2.14**	-2.16**	-1.07**	
MX1	+1.43*	-0.44	-1.87**	+0.47	+1.24*	+0.54	+1.97**	+1.54**	+1.01*	
PADI4	+0.37	-1.75**	-2.11**	+0.13	+0.42	+0.12	+2.32**	+2.07**	+1.95**	
MPO	-0.48*	+2.07*	+2.56**	-0.19	-0.57*	-1.27	-2.13*	-2.18**	-0.91*	

d NETosis Assay (Elastase Release)



e Neutrophil RNA response programs

	Ctrl FB vs alone	UC FB vs alone	UC vs Ctrl FB	α5β1i alone	α5β1i in UC FB
Inflammatory cytokines	+0.32*	+0.40***	+0.08	+0.35***	-0.07
Interferon response	+0.06*	+0.04	-0.01	+0.06*	+0.01
Granule/protease	+0.02	+0.02	-0.02	+0.07	+0.06***
NADPH oxidase	+0.25**	+0.28**	-0.02	+0.28***	-0.05
Chemotaxis	+0.46***	+0.48***	-0.00	+0.50***	-0.14*
Hypoxia/glycolysis	+0.19**	+0.14***	-0.06	+0.20***	+0.01

f Predicted fibroblast–neutrophil routes

	UC vs Ctrl FB	α5β1i vs UC FB
1 COL1A2 → CD44 ECM-FB → OSM	-0.79	+0.82
2 FN1 → CD44 ECM-FB → CXCR4	-0.64	+0.67
3 IL1B → IL1R2 CCN-FB → MX1/ISG	-0.45	+0.62
4 CEACAM1 → CEACAM8 ECM-FB → PADI4	+0.41	-0.41
5 CXCL12 → CXCR4 CCN-FB → CXCR4	+0.65	-0.65
6 C5 → C5AR1 ECM-FB → CXCR4	+0.79	-0.79

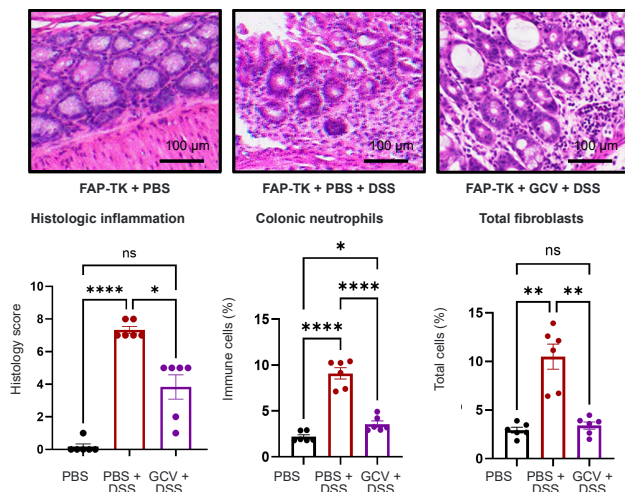
LR consensus change (−0.9 to +0.9)

g Module support for ligands in f

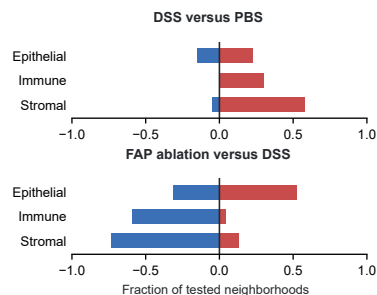
	Inflam. cytokines	IFN response	Granule/protease	NADPH oxidase	Chemo-taxis	Hypoxia/glycolysis
1 COL1A2	5	6	2	1	1	8
2 FN1	5	10	2	1	1	7
3 IL1B	8	10	3	0	4	2
4 CEACAM1	5	10	2	2	2	7
5 CXCL12	7	10	2	2	1	7
6 C5	8	10	2	1	1	7

Matched ligands; color: prior rank; dot size/count: top decile

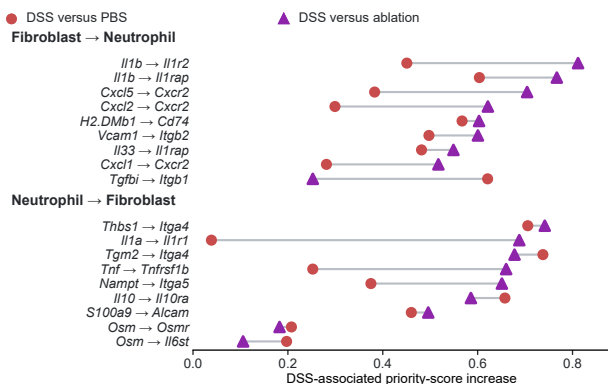
a FAP ablation reduces DSS-colitis pathology



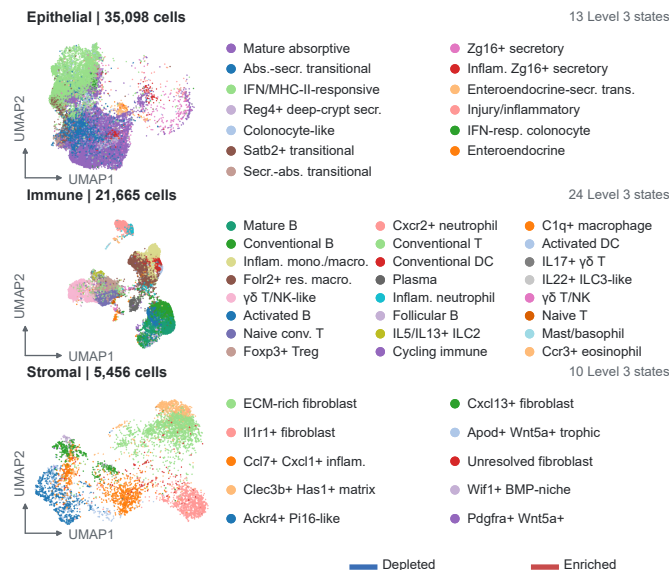
c MitoR neighborhood remodeling



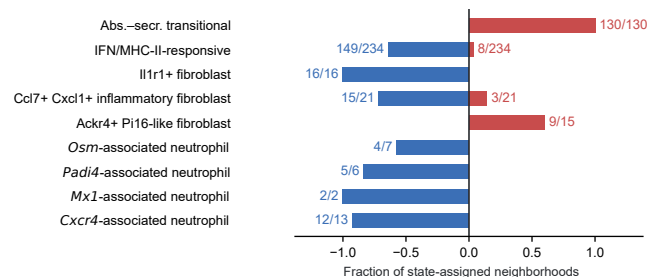
d Reciprocal signalling routes



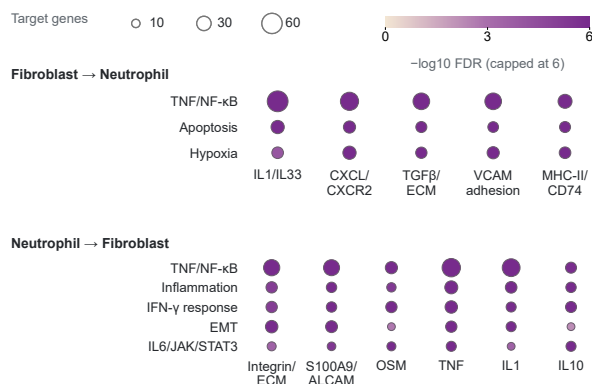
b Compartment-specific UMAPs



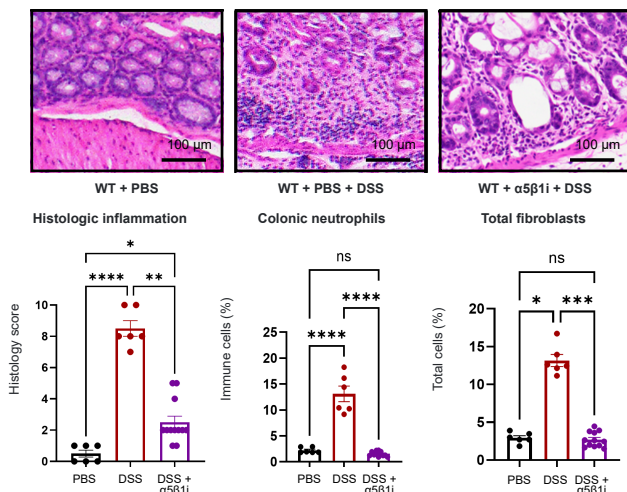
Level 3 states | intervention versus DSS



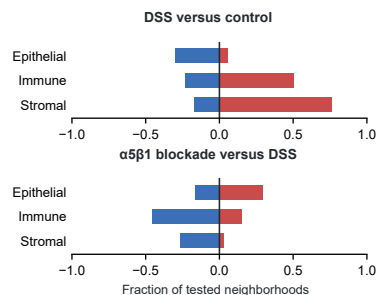
e Predicted receiver-cell pathways



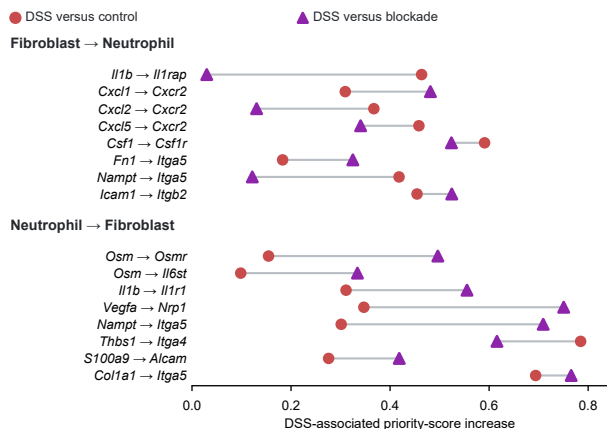
a α5β1 blockade reduces DSS-colitis pathology



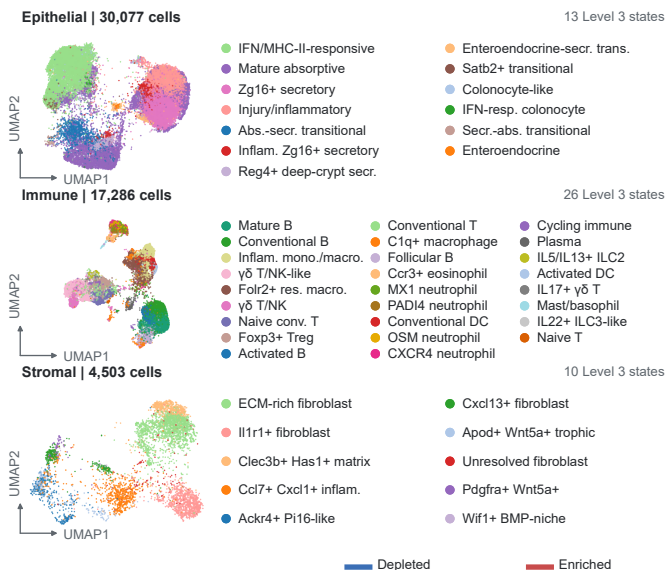
c MitoR neighborhood remodeling



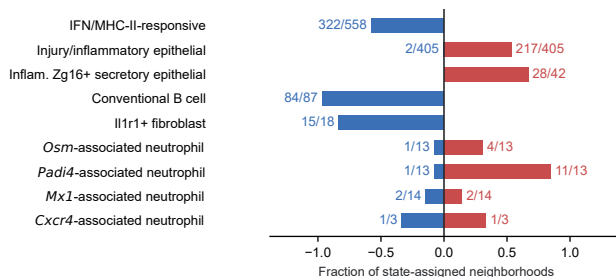
d Reciprocal signaling routes



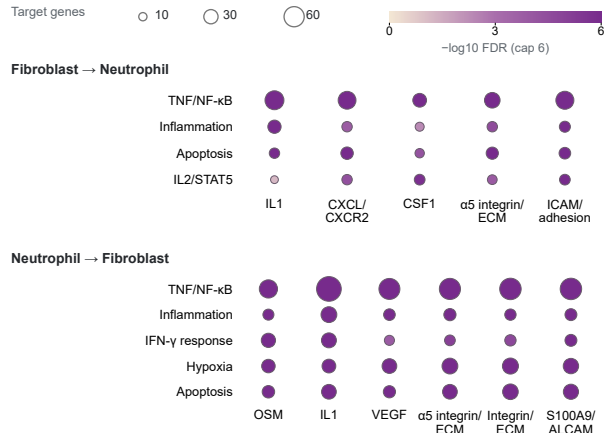
b Compartment-specific UMAPs



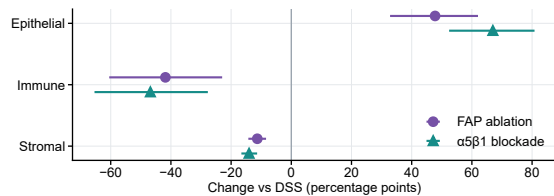
Level 3 states | intervention versus DSS



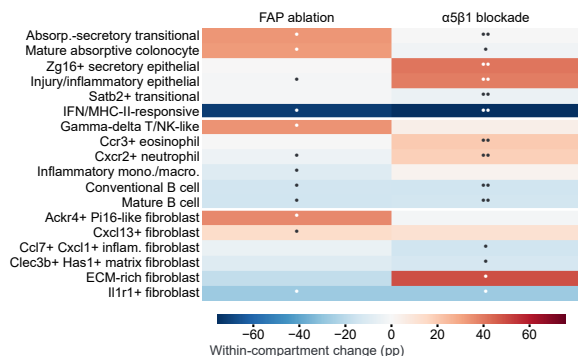
e Predicted receiver-cell pathways



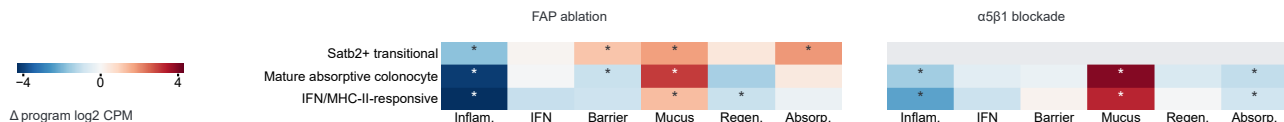
a



c

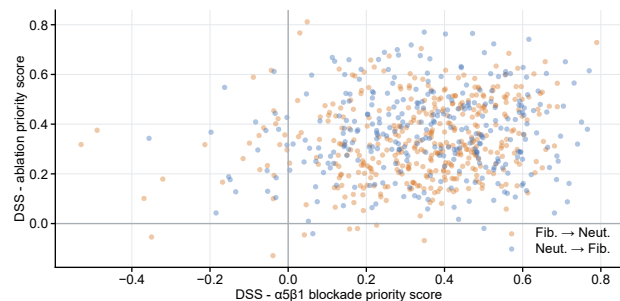


e

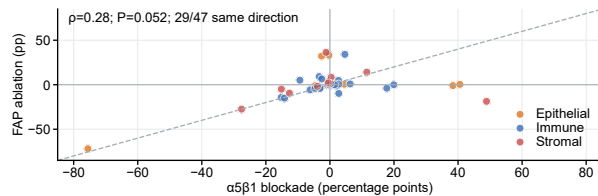


f

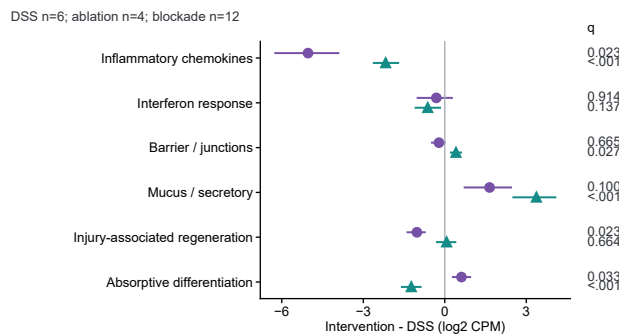
691 routes: $\rho=0.15$, $P=0.00013$; core: 490/502 attenuated



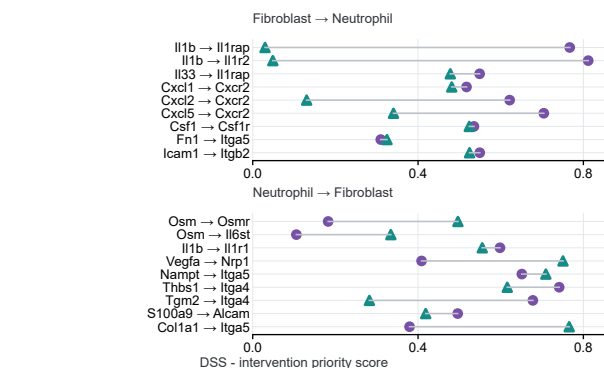
b



d

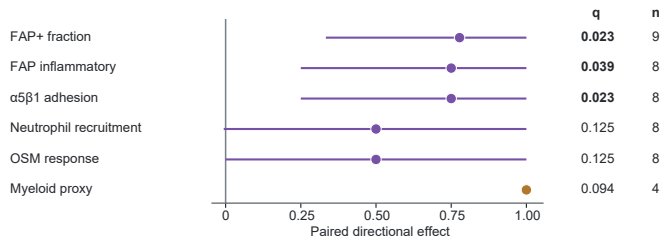


g



a TAURUS UC: paired inflammation

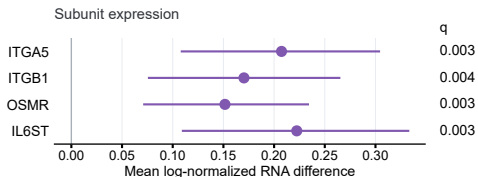
Inflamed vs noninflamed; 95% bootstrap CI



Myeloid: monocyte proxy; TAURUS did not retain neutrophils.

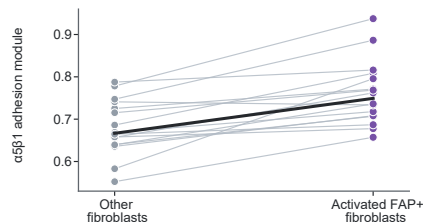
c Receptor availability in activated FAP+ fibroblasts

Activated FAP+ minus other fibroblasts; 16 paired patients; 50 matched biopsies



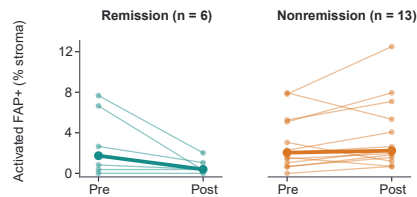
b FAP+ fibroblast adhesion state

16 patients; median paired Δ = 0.07



d Activated FAP+ cells after anti-TNF

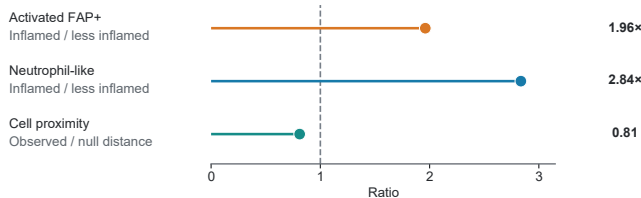
Site-matched pre/post adalimumab



Lines: patients; bold lines: group medians

e Paired spatial validation in UC

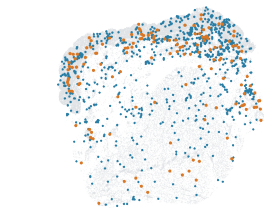
One donor; paired UC sections (SCP3818)



Proximity: P = 0.005; 1,000 size-matched null sets

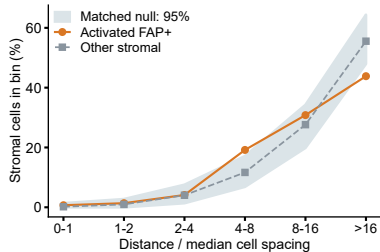
f Spatial context of the activated-FAP niche

UC1 inflamed; one donor



● Neutrophil-like (n=879)
● Activated FAP+ (n=146)

Nearest neutrophil-like cell



g Cross-system pathway responses

